

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. Examination (EC)

November 2013

EC 208 Analog Electronic Circuits

Date: 14.11.2013, Thursday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) Compare power amplifier classes in regard to their operating efficiency [02]
(b) Explain Class A Amplifier Operation [05]
- Q - 2 (a) Describe quiescent conditions and formation of sweep in bootstrap time base generator with necessary waveforms. [07]
(b) Draw and explain Flash type analog to digital converter [07]

OR

- Q - 2 (a) Explain the piezoelectric effect. With a sketch of equivalent circuit explain Crystal Oscillator. [07]
(b) Describe the operation of Emitter Coupled Clipper [07]
- Q - 3 (a) With a sketch of block diagram, describe successive approximation type Analog to Digital Converter [07]
(b) With the help of the V – I characteristic and load line, explain (i) the monostable and (ii) the bistable operations of current controlled negative resistance devices [07]

OR

- Q - 3 (a) Describe Transistor Clipper with illustrations. [07]
(b) Explain the operation of Tunnel Diode illustrating its V-I Characteristics [07]

SECTION – II

- Q - 4 (a) What is the use of positive and negative feedback for an amplifier. [02]
(b) Enlist the features of time base signal [03]
- Q - 5 (a) Draw the block diagram of Voltage Series and Voltage Shunt Feedback configuration for an amplifier [06]

- (b) Draw the circuit diagram of series and shunt voltage regulator. Explain the functioning of differential amplifier. [06]

OR

- Q - 5 (a) Describe transistor constant current sweep with necessary circuit diagrams and output waveforms [06]

- (b) Describe IC Voltage Regulators. [06]

Q - 6 Attempt Any Three [18]

1. Through help of any one example, describe biased Clipping at two independent levels
2. Describe the meaning of current controllable negative resistance (NR) device and voltage controlled negative resistance device. Give at least three examples of both types of NR Devices
3. Describe R-2R Ladder type Digital to Analog conversion with necessary calculations and output diagrams
4. Describe the weighted register type Digital to Analog Convert using circuit diagram.